

Roll No.....

Mid Semester Examination 2025

Name of the Program: B.Tech.

Semester: V

Name of the Course: Digital Signal Processing

Course Code: TEC501

Time: 90 minutes

Maximum Marks: 50

Note:

- (i) Answer **all the questions** by choosing *any one of the sub questions*.
(ii) Each question carries 10 marks

Q1	(10 marks)	CO1
(a)	Explain the basic elements Digital Signal Processing system. Point the advantage of digital over analog signal processing.	
OR		
(b)	a. Determine the periodic signal $x(n)$, with fundamental period $N=8$, if their Fourier Coefficients are given by: $C_k = \begin{cases} \sin\left(\frac{k\pi}{n}\right) & ; 0 \leq k \leq 6 \\ 0; & k = 7 \end{cases}$	
Q2	(10 marks)	CO1
(a)	Illustrate the discrete time sinusoid properties with proof.	
OR		
(b)	Determine the Energy and Power of the following sequence. i) $x(n) = u(n)$ ii) $x(n) = (-0.5)^n \cdot u(n)$	
Q3	(10 marks)	CO1
(a)	Explain “Cross Correlation” and “Auto- Correlation” Functions. Compute the ‘Cross Correlation’ and ‘Linear Convolution’ of the following sequence: $\begin{matrix} x_1(n) = \{1, 2, 3, 4\} & \text{and} & x_2(n) = \{2, 3, 1, 1\} \\ \uparrow & & \uparrow \end{matrix}$	
OR		
(b)	i) State and prove the “Parseval’s Relation” for discrete time aperiodic sequence. ii) Determine DTFT and energy density spectrum (ESD) of a sequence as: $x(n) = \begin{cases} a^n u(n), & a < 1 \\ 0, & \text{otherwise} \end{cases}$	
Q4	(10 marks)	CO2
(a)	Use the ‘DFT as Linear Transformation Technique’ to compute 4- point DFT of the sequence $x(n) = \{0, 1, 2, 3\}$ $\uparrow$ Compare the computational complexity of DFT and FFT algorithm.	

OR		
(b)	Proof that the multiplication of two DFTs is equivalent to circular convolution.	
Q5	(10 marks)	
(a)	Obtain and prove the 8-point butterfly flowgraph for DIFFFT (Decimation-in-Frequency- Fast Fourier Transform).	CO2
OR		
(b)	Compute the Inverse DFT for the following using FFT algorithm: $X(k) = \{20, -5.828-j2.414, 0, -0.172-j0.414, 0, -0.172+j0.414, 0, -5.828+j2.414\}$ ↑	